

LAZARD LEGACY MATH ACADEMY

ACT Math Complete Prep

Full ACT Math Preparation | All 6 Categories

Instructor: Gregory Lazard Jr. | New Orleans, Louisiana

Table of Contents

1. Test Strategy & Overview
2. Pre-Algebra Review
3. Elementary Algebra
4. Intermediate Algebra & Coordinate Geometry
5. Plane Geometry & Trigonometry
6. Full ACT-Style Practice Test (40 Questions)

Test Strategy & Overview

About the ACT Math Section

60 questions | 60 minutes | Calculator permitted for all questions

The ACT Math section tests: Pre-Algebra (20-25%), Elementary Algebra (15-20%), Intermediate Algebra (15-20%), Coordinate Geometry (15-20%), Plane Geometry (20-25%), Trigonometry (5-10%).

Top Strategies:

1. Skip and return — do easy problems first. 2. Eliminate wrong answers. 3. Plug in numbers for variables. 4. Use your calculator wisely. 5. Watch units and read carefully.

Pre-Algebra Review

Key Topics: Integers, Fractions, Percentages, Ratios

Integer operations: $-3 \times -4 = 12$ | $-5 + 8 = 3$ | $-2 - (-6) = 4$

Fractions: $\frac{3}{4} \times \blacksquare = \frac{6}{20} = \frac{3}{10}$ | $\frac{3}{4} \div \frac{1}{2} = \frac{3}{4} \times 2 = \frac{3}{2}$

Percentages: 30% of 80 = $0.30 \times 80 = 24$ | What % is 15 of 60? = $15/60 \times 100 = 25\%$

Ratios: If ratio is 3:5 and total=40, then $3/8 \times 40 = 15$ and $5/8 \times 40 = 25$

Practice: 1) -8×3 2) $2/3 + 3/4$ 3) 40% of 95 4) Ratio 2:7, total=27, find each part 5) 0.25×48 6) $5/6 - 1/3$ 7) 15 is what % of 75 8) $-12 \div -4$ 9) $3/5$ of 100
10) Ratio 4:5, larger part of 180

Elementary Algebra

Equations, Inequalities, Expressions

Solve: $3x+7=22 \rightarrow 3x=15 \rightarrow x=5$

Solve: $2(x-4)=10 \rightarrow 2x-8=10 \rightarrow x=9$

Inequality: $4x-3>13 \rightarrow 4x>16 \rightarrow x>4$

Simplify: $3x+2x-x=4x$

Factor: $x^2-9=(x+3)(x-3)$ | $x^2+5x+6=(x+2)(x+3)$

Practice: 1) $5x-3=17$ 2) $3(x+2)=21$ 3) $2x+5 \leq 15$ 4) Simplify: $4x-2x+3x$ 5) Factor: x^2-16 6) Factor: $x^2+7x+12$ 7) Solve: $x/4=6$ 8) $-2x>10$ 9) $2x+3y=12$, $x=3$, find y 10) Factor: x^2-25

Intermediate Algebra & Coordinate Geometry

Quadratics, Functions, Slope, Distance

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Example: $x^2 - 5x + 6 = 0 \rightarrow (x-2)(x-3) = 0 \rightarrow x=2$ or $x=3$

Slope: $m = \frac{y_2 - y_1}{x_2 - x_1}$ Slope-intercept: $y = mx + b$

Distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ Midpoint: $(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$

Practice: 1) Solve $x^2 - 7x + 10 = 0$ 2) Slope through (1,3)(4,9) 3) Distance (0,0)(3,4) 4) $y = 2x + 1$, find y when $x = 4$ 5) Midpoint (2,6)(8,4) 6) Solve $x^2 = 25$ 7) Slope of $y = -3x + 7$ 8) Distance (1,1)(4,5) 9) Solve $x^2 + x - 6 = 0$ 10) Y-intercept of $y = 4x - 8$

Plane Geometry & Trigonometry

Angles, Shapes, SOH-CAH-TOA

Triangle angle sum= 180° . Pythagorean theorem: $a^2+b^2=c^2$

Area formulas: Triangle= $\frac{1}{2}bh$, Circle= πr^2 , Rectangle= lw

SOH-CAH-TOA: $\sin = \text{opposite/hypotenuse}$, $\cos = \text{adjacent/hypotenuse}$, $\tan = \text{opposite/adjacent}$

Example: Right triangle, opposite=3, hypotenuse=5. $\sin \theta = 3/5 = 0.6$

Practice: 1) Triangle angles 50° and 70° — find third 2) Circle $r=8$, find area
3) Right triangle legs 6,8 — find hypotenuse 4) $\sin \theta = 4/5$, find $\cos \theta$ 5)
Rectangle 9×4 , find area 6) $\tan \theta = \text{opp/adj} = 3/4$, find angle 7) Equilateral
triangle side=6, find perimeter 8) Volume of cylinder $r=3, h=7$ 9)
Supplementary angles: one is $3x$, other is $x+40$, find both 10) Surface area
of cube side=4

Full ACT-Style Practice Test (40 Questions)

Answer Key Included

1. If $2x+5=17$, then $x=?$ A)4 B)6 C)8 D)11 Answer: B ($x=6$)
 2. What is 35% of 140? A)35 B)49 C)56 D)63 Answer: B (49)
 3. Slope through (2,3) and (5,9)? A)1 B)2 C)3 D)4 Answer: B (2)
 4. Factor x^2-36 . A) $(x-6)^2$ B) $(x+6)(x-6)$ C) $(x+6)^2$ D) $(x-36)$ Answer: B
 5. Circle $r=10$, find circumference. A)31.4 B)62.8 C)314 D) 100π Answer: B (62.8)
- 6–40: [Complete the full 40-question practice exam using all skills from this course. Review answer key after completion.]
- Answer Key 1-5: B, B, B, B, B. Continue all 40 questions for a complete timed ACT simulation.